# Unit 2

# Introduction to Computer Networks

# [5 Hrs]

*2.1 Definitions, Uses, Benefits*

*2.2 Network Topologies and its types (Bus, Star, Ring, Mesh, Hybrid)*

*2.3 Types of Computer Networks (PAN, LAN, MAN, WAN, CAN)*

*2.4 Networking Types (Client/Server, P2P)*

*2.5 Overview of Protocols and Standards*

*2.6 OSI Reference Model*

*2.7 TCP/IP Models and its Comparison with OSI*

*2.8 Networking Hardware: NIC, Hub, Repeater, Switch, Bridge, Router*

*2.9 Basic Networking Commands*

**Sanjeev Thapa, Er. DevOps, SRE, CKA, RHCSA, RHCE, RHCSA-Openstack, MTCNA, MTCTCE, UBSRS, HEv6, Research Evangelist**

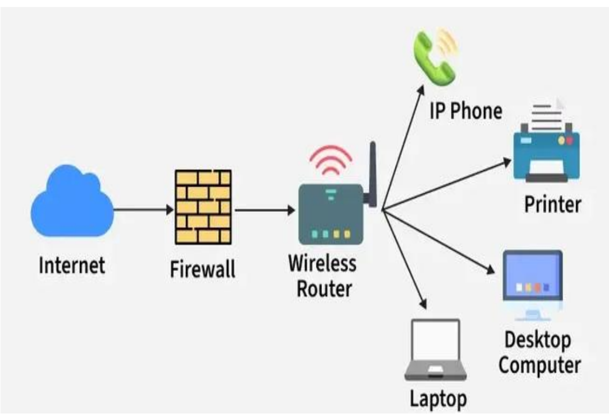
# 2.1 Definitions, Uses, Benefits

## Definition of Computer Network

A **computer network** is a collection of two or more computers or devices connected together to share data, resources, and services.

These devices may be connected using:

- cables
- fiber optics
- Wi-Fi
- radio waves




When computers communicate and share resources with each other, it is called a **computer network**.

**Examples of a Computer Network**

- computers connected in a school lab
- office computers sharing one printer
- mobile phone using Wi-Fi internet
- ATM network of banks
- hospital computers sharing patient records

## Uses of Computer Networks

Computer networks are used for many purposes:

**1. Resource Sharing**

Devices can share:

- printer
- scanner
- storage
- internet connection

**Example:**

In a college office, many computers may use one printer through a network.

**2. Data Sharing**

Users can send and receive:

- files
- documents
- photos
- videos

**Example:**

A teacher sends notes to students through the college network.

**3. Communication**

Networks help people communicate through:

## Benefits of Computer Networks

**1. Cost Saving**

Instead of buying separate printers, storage, and internet for each computer, devices can share them.

**2. Easy Communication**

Users can exchange information quickly.

**3. Better Data Management**

Data can be stored centrally and managed efficiently.

**4. Increased Reliability**

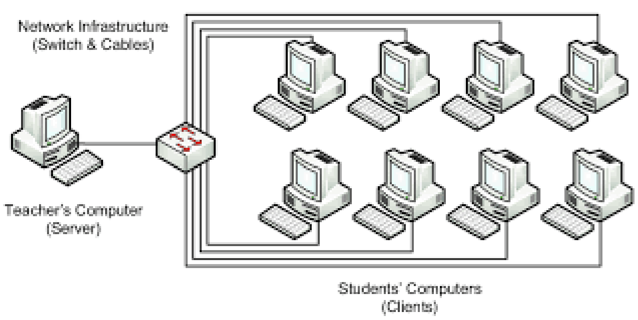
Backup systems and multiple connections can improve reliability.

**5. Scalability**

New devices can be added easily.

**6. Collaboration**

Many users can work together on project




- email
- chat
- video call
- voice call

**Example:**

A team meeting can be done using Zoom or Google Meet over a network.

**4. Centralized Management**

Data and software can be controlled from one central server.

**Example:**

In a bank, all branch computers may connect to a central database server.

**5. Remote Access**

Users can access information from distant places.

**Example:**

A person in Butwal can access a server in Kathmandu through the internet.

## 2.2 Network Topologies

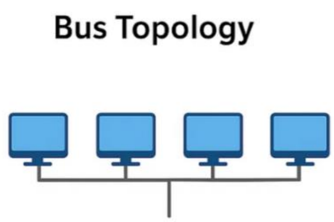
**Network topology** means the physical or logical arrangement of computers and network devices in a network.

In simple words, it shows **how devices are connected**.

### 1. Bus Topology

In **bus topology**, all devices are connected to a single main cable called the **backbone cable**.

**Structure:**




All computers share one communication line.

**Example:**

A small old office network where all computers connect to one long cable.

**Advantages:**

- simple to install
- low cost
- less cable required

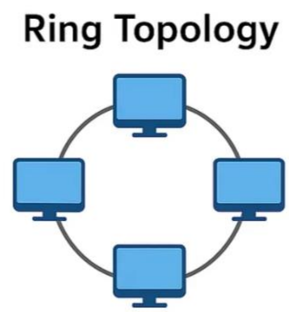
**Disadvantages:**

- if main cable fails, whole network fails
- difficult to troubleshoot
- performance decreases with more devices

### 3. Ring Topology

In **ring topology**, each device is connected to two neighboring devices, forming a circular path.

**Structure:**




Data travels around the ring.

**Example:**

Older token ring networks.

**Advantages:**

- orderly data transmission
- less collision

**Disadvantages:**

- failure in one node/cable may affect whole network
- difficult to reconfigure

### 4. Mesh Topology

In **mesh topology**, every device is connected to every other device.

**Structure:**

Multiple paths exist between devices.

**Example:**

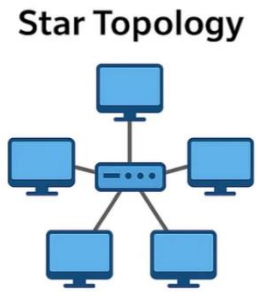
Critical military or research communication systems.

**2. Star Topology**

In **star topology**, all devices are connected to a central device such as a **hub** or **switch**.

**Structure:**

Each computer has a separate cable to the central device.




**Example:**

Most modern school and office LANs use star topology.

**Advantages:**

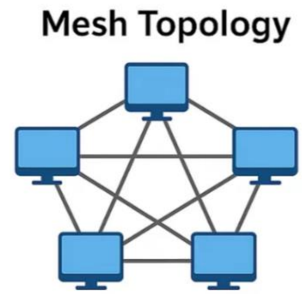
- easy to install and manage
- failure of one cable affects only one device
- easy to expand

**Disadvantages:**

- if central device fails, entire network fails
- needs more cable

**Advantages:**

- very reliable
- high fault tolerance
- secure communication




**Disadvantages:**

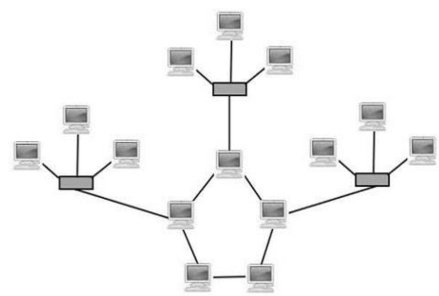
- very expensive
- large amount of cabling
- difficult installation

**5. Hybrid Topology**

A **hybrid topology** is a combination of two or more topologies.

**Example:**

A college network may use star topology inside each department and combine departments in another form.



**Advantages:**

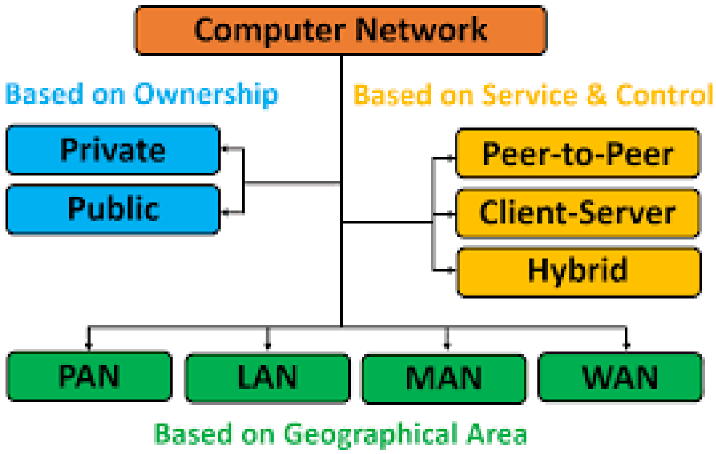
- flexible
- scalable
- can combine strengths of different topologies

**Disadvantages:**

- complex design
- higher cost

# 2.3 Types of Computer Networks

Computer networks can be classified according to their **size and coverage area.**




### 1. PAN – Personal Area Network

A **PAN** is a small network around one person.

**Range:**


Very short distance, usually a few meters.

**Examples:**

- Bluetooth connection between phone and earbuds
- phone connected to laptop hotspot

**Use:**

Personal devices communication.

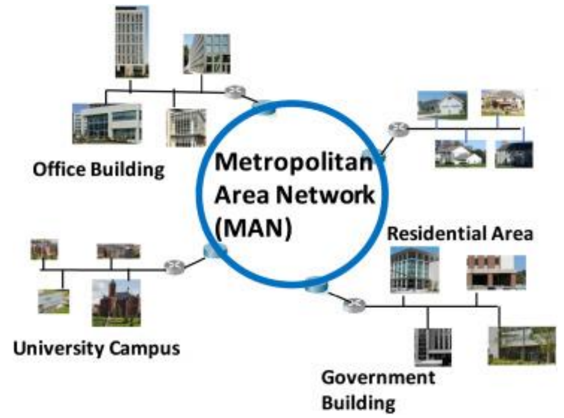
### 2. LAN – Local Area Network

A **LAN** connects computers within a small area such as:

- room

### 3. MAN – Metropolitan Area Network

A **MAN** covers a city or town.




**Example:**

A network connecting different offices of a company across Butwal city.

**Features:**

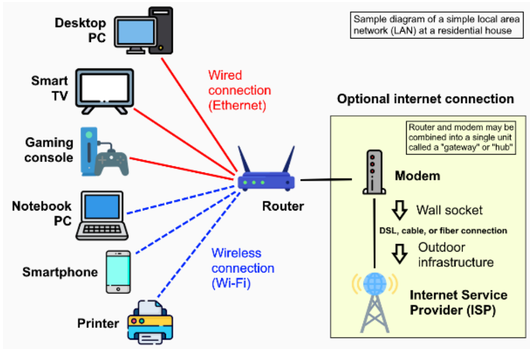
- larger than LAN
- smaller than WAN


### 4. WAN – Wide Area Network

A **WAN** covers a large geographical area such as:

- country
- continent
- world

- home
- office
- school lab




**Example:**

A computer lab in a college.

**Features:**

- high speed
- private ownership
- low cost

**Example:**

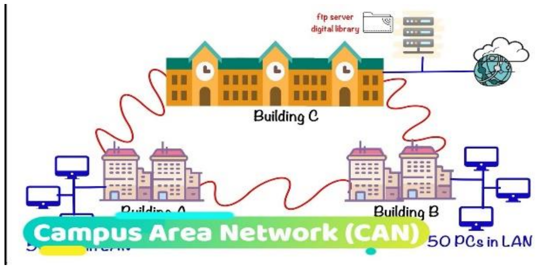
The internet is the best example of WAN.

**Features:**

- connects distant networks
- may use leased lines, fiber, satellite

**5. CAN – Campus Area Network**

A **CAN** connects multiple LANs within a campus or institution.




**Example:**

A university network connecting administration block, library, labs, and hostels.

**Features:**

- larger than LAN
- smaller than MAN

common in colleges and hospitals

## 2.4 Networking Types (Client/Server, P2P)

### 1. Client/Server Network

In a **client/server network**, one or more computers act as **servers**, and others act as **clients**.

**Server**

A server provides:

- files

### 2. Peer-to-Peer (P2P) Network

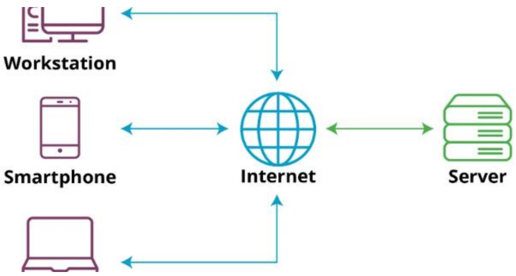
In a **peer-to-peer network**, all computers are equal. Each computer can act as both client and server.

**Example:**

Two computers sharing files directly in a small office.

**Advantages:**

- database
- printer service
- web service




**Client**

A client requests services from the server.

**Example:**

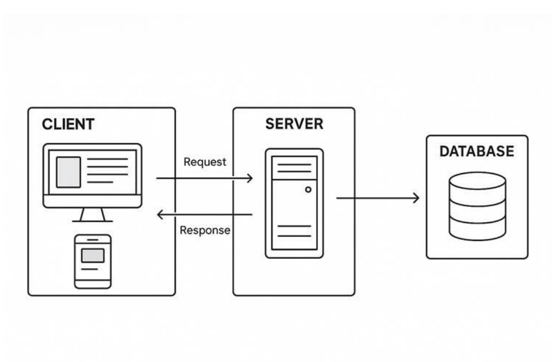
In a bank, employees' computers act as clients and connect to a central database server.

**Advantages:**

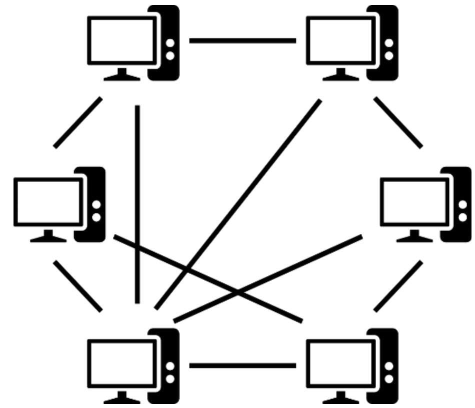
- centralized control
- better security
- easy backup
- efficient management

**Disadvantages:**

- expensive
- server failure affects many users
- requires administration

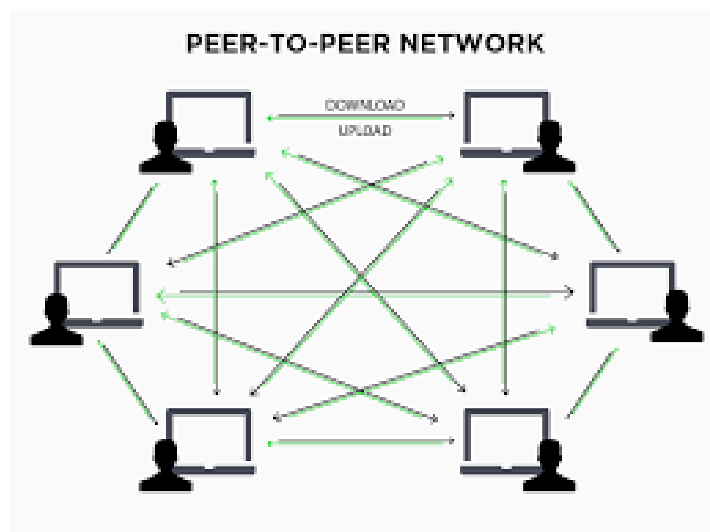



- simple
- low cost
- no dedicated server needed



**Disadvantages:**

- less security
- difficult management
- not suitable for large networks



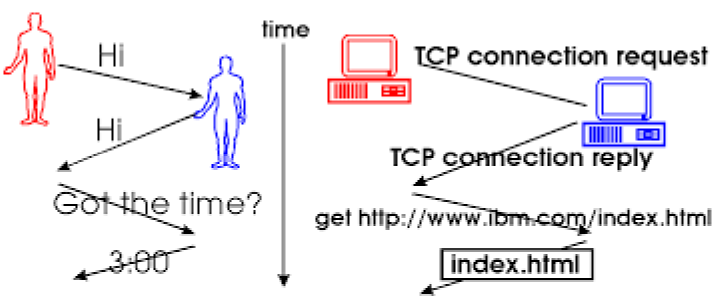
## 2.5 Overview of Protocols and Standards

**Protocol**

A **protocol** is a **set of rules used for communication between devices.**

Protocols define:

- how data is formatted
- how it is sent
- how it is received
- how errors are handled




Protocols are rules for communication between network devices. Standards ensure compatibility among devices and technologies. Both are necessary for successful computer networking.

**Examples:**

- HTTP
- HTTPS
- FTP
- TCP
- IP
- SMTP

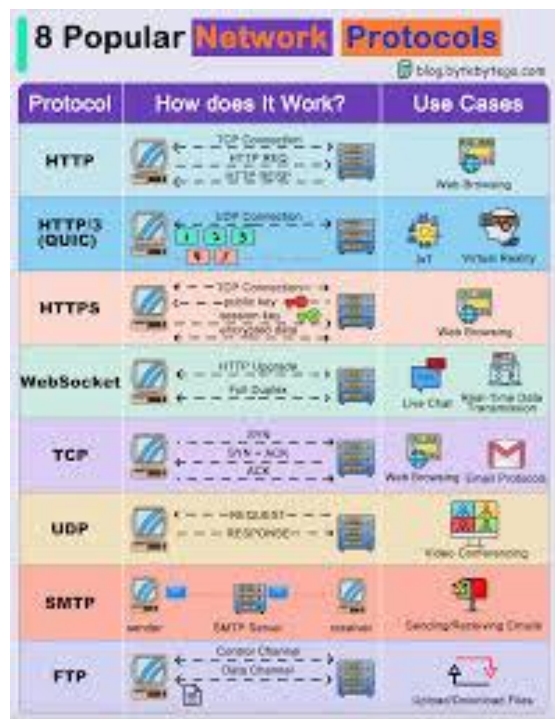
**Why Protocols and Standards are Important**

- ensure communication between different devices
- make networking reliable
- support interoperability
- reduce confusion
- improve compatibility

**Examples of Common Protocols**

**HTTP**

Used for web browsing.




**HTTPS**

Secure version of HTTP.

**FTP**

Used for file transfer.

**SMTP**

Used for sending email.

**TCP**

Provides reliable data delivery.

**IP**

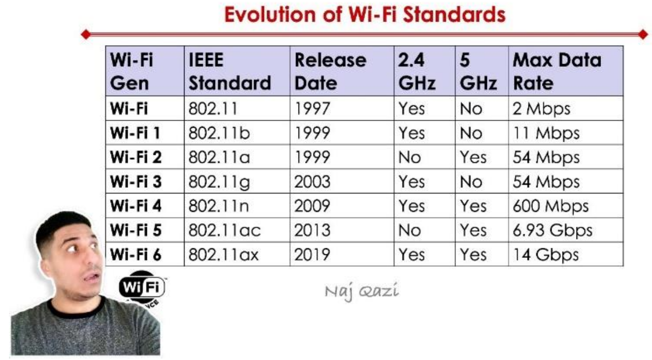
Handles addressing and routing.

**Standard**

A **standard** is an agreed rule or guideline followed by manufacturers and organizations to ensure compatibility.

**Example:**

Wi-Fi standards allow devices from different companies to connect.


Evolution of Wi-Fi Standards

| Wi-Fi Gen | IEEE Standard | Release Date | 2.4 GHz | 5 GHz | Max Data Rate |
|---|---|---|---|---|---|
| Wi-Fi | 802.11 | 1997 | Yes | No | 2 Mbps |
| Wi-Fi 1 | 802.11b | 1999 | Yes | No | 11 Mbps |
| Wi-Fi 2 | 802.11a | 1999 | No | Yes | 54 Mbps |
| Wi-Fi 3 | 802.11g | 2003 | Yes | No | 54 Mbps |
| Wi-Fi 4 | 802.11n | 2009 | Yes | Yes | 600 Mbps |
| Wi-Fi 5 | 802.11ac | 2013 | No | Yes | 6.93 Gbps |
| Wi-Fi 6 | 802.11ax | 2019 | Yes | Yes | 14 Gbps |

Naj Qazi

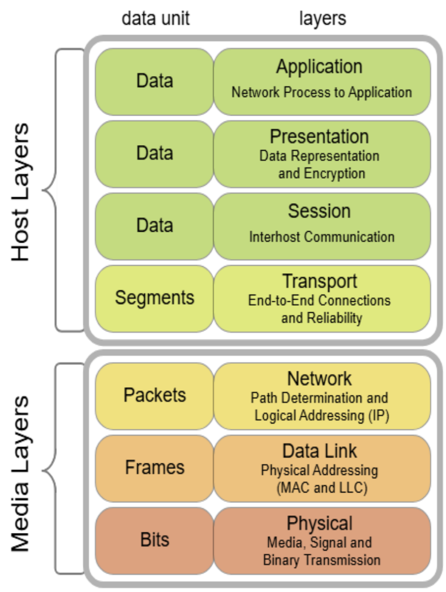
## 2.6 OSI Reference Model

The **OSI Model** stands for **Open Systems Interconnection Model**.
It is a conceptual framework that explains how data moves from one device to another through a network.

It has **7 layers**.

The OSI reference model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application.

Each layer has a specific function and helps in structured network communication.




### 1. Physical Layer

Responsible for transmitting raw bits through the medium.

**Functions:**

- cables

### 4. Transport Layer

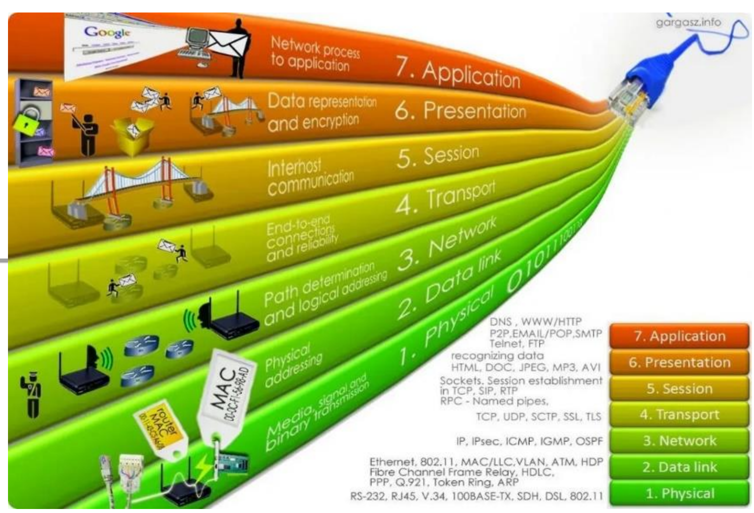
Responsible for end-to-end delivery.

**Functions:**

- segmentation
- flow control
- error control

**Example:**

TCP, UDP




### 5. Session Layer

- connectors
- signal transmission

**Example:**

Ethernet cable, hub

**2. Data Link Layer**

Responsible for node-to-node delivery and error detection.

**Functions:**

- framing
- MAC addressing
- error detection

**Example:**

Switch, NIC

**3. Network Layer**

Responsible for logical addressing and routing.

**Functions:**

- IP addressing
- routing

**Example:**

Router

Responsible for establishing, managing, and terminating sessions.

**Example:**

Login session between systems.

**6. Presentation Layer**

Responsible for data translation, encryption, and compression.

**Example:**

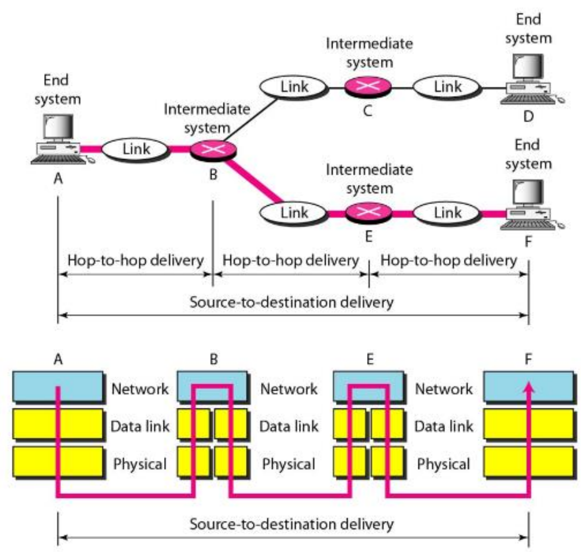
Data format conversion, SSL

**7. Application Layer**

Closest to the user; provides network services to applications.

**Examples:**

HTTP, FTP, SMTP, DNS



# 2.7 TCP/IP Models and its Comparison with OSI

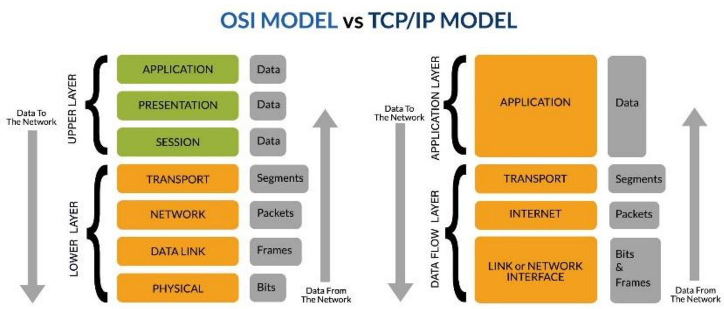


### Introduction

The **TCP/IP model** is a networking model used in real computer networks, especially on the **Internet**.
It explains how data moves from one device to another.

**TCP/IP** stands for:

- **TCP** = Transmission Control Protocol
- **IP** = Internet Protocol




The TCP/IP model is simpler than the OSI model and has **4 layers**.

### Layers of TCP/IP Model

### 1. Application Layer

This layer provides services directly to user applications.

It combines the functions of:

- Application layer
- Presentation layer
- Session layer
  of the OSI model.

**Examples of protocols:**

- HTTP
- HTTPS

### 2. Transport Layer

This layer is responsible for end-to-end communication between source and destination.

**Examples of protocols:**

- TCP
- UDP

**Functions:**

- segmentation
- flow control
- error control
- reliable delivery

### 3. Internet Layer

This layer is responsible for logical addressing and routing.

**Example of protocol:**

- IP

**Functions:**

- IP addressing
- packet routing
- path selection

### 4. Network Access Layer

This is the lowest layer of TCP/IP model.
It handles actual transmission of data through the physical network.

- FTP
- SMTP
- DNS

**Functions:**

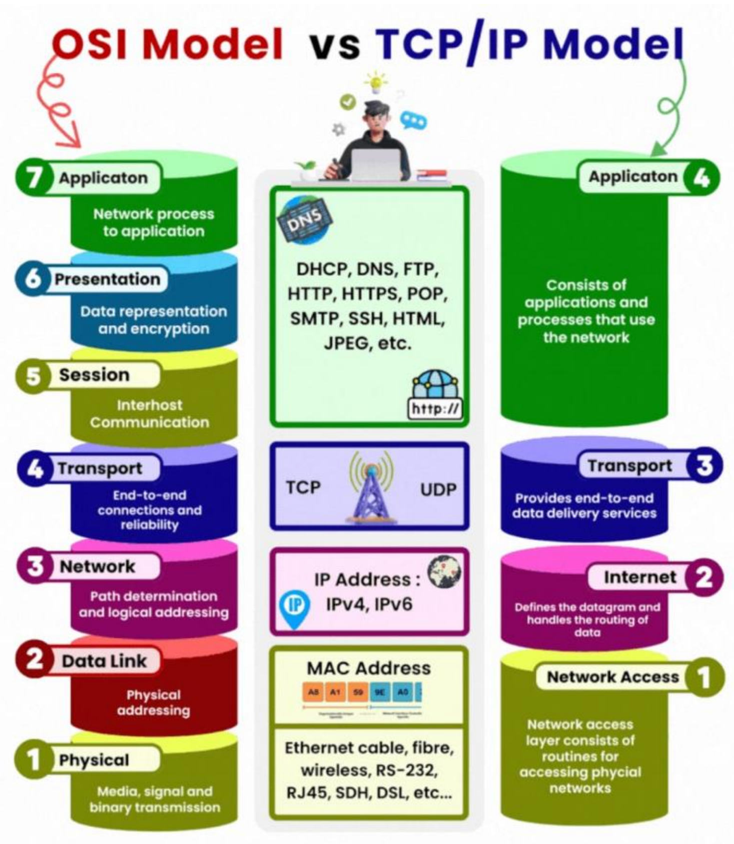
- file transfer
- email
- web browsing
- remote login

It combines the functions of:

- Data Link layer
- Physical layer
  of the OSI model.

**Functions:**

- framing
- MAC addressing
- transmission of bits
- error detection at local link




## Comparison Between TCP/IP and OSI

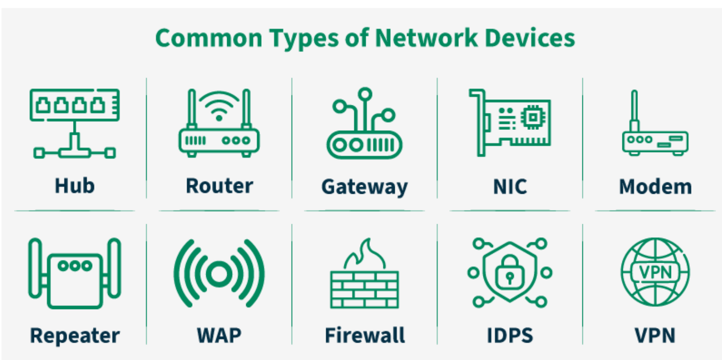
| **OSI Model** | **TCP/IP Model** |
|---|---|
| 7 layers | 4 layers |
| Theoretical model | Practical model |
| Developed by ISO | Developed by DARPA |
| Session and Presentation are separate | Session and Presentation are included in Application layer |
| Physical and Data Link are separate | Combined as Network Access layer |
| Used for understanding network concepts | Used in real Internet communication |

# 2.8 Networking Hardware:

Networking hardware are the physical devices used to build and operate a computer network. These devices help computers communicate, send data, receive data, and connect different parts of a network.

**The main networking hardware are:**

- NIC
- Hub
- Repeater
- Switch
- Bridge
- Router




---

## 1. NIC (Network Interface Card)

A **NIC** is a hardware component that allows a computer or device to connect to a network.

It may be:

- wired NIC
- wireless NIC

**Functions:**

- connects computer to network
- provides physical address called **MAC address**
- sends and receives data

**Example:**

- Ethernet card in desktop computer
- Wi-Fi adapter in laptop

**Simple example:**

If a computer wants to connect to the internet through LAN cable, it needs a NIC.

---

## 2. Hub

A **hub** is a simple networking device that connects multiple computers in a network.

When a hub receives data from one device, it sends that data to **all connected devices**.

**Functions:**

- connects multiple devices
- broadcasts data to all ports

**Advantages:**

- simple
- low cost

**Disadvantages:**

- sends data to all devices
- causes more traffic
- less secure
- less efficient

**Example:**

Old small office networks used hubs to connect computers.

---

## 3. Repeater

A **repeater** is a device that regenerates or strengthens weak signals.

When data travels a long distance, the signal becomes weak. A repeater refreshes the signal and sends it again.


**Functions:**

- receives weak signal
- regenerates signal
- extends transmission distance

**Example:**

If a LAN cable is very long, a repeater can be used to extend the network.

---

## 4. Switch

A **switch** is a networking device that connects multiple computers in a LAN and sends data only to the correct destination device.

Unlike a hub, a switch does not broadcast to all devices unnecessarily.

**Functions:**

- connects multiple devices
- forwards data to specific device
- uses MAC address table

**Advantages:**

- faster than hub
- less traffic
- more secure
- better performance

**Example:**

Modern computer labs and offices use switches.

**Simple example:**

If computer A sends data to computer B, the switch sends it only to B, not to all computers.

---

## 5. Bridge

A **bridge** is a device that connects two LAN segments and filters network traffic.

It works at the data link layer and uses MAC addresses.


**Functions:**

- connects two parts of a network
- reduces unnecessary traffic
- filters data

**Example:**

If one office network is divided into two parts, a bridge can connect them.

**Advantage:**

- reduces collision
- improves performance

---

### 6. Router

A **router** is a networking device that connects different networks and routes data packets from one network to another.

It works using **IP addresses**.

**Functions:**

- connects different networks
- chooses best path for data
- routes packets
- connects LAN to internet

**Example:**

Home Wi-Fi router connects your home network to the internet.


**Simple example:**

If your laptop accesses a website, the router forwards your request from local network to the internet.

## 2.9 Basic Networking Commands

Basic networking commands are used to:

**WIN + R , CMD**

- check network connection
- test communication between devices
- view IP address
- troubleshoot internet problems
- check routes, DNS, and ports

These commands are very useful for students, network administrators, and computer users.

---

### A. Basic Networking Commands in Windows

#### 1. ipconfig

This command shows the basic IP configuration of the computer.

**Use:**

- view IP address
- view subnet mask
- view default gateway

**Command:**

ipconfig

**Example output:**

- IPv4 Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

**Purpose:**

Used to know the current network settings of the computer.

---

## 2. ipconfig /all

This shows detailed network information.

**Command:**

ipconfig /all

**Shows:**

- IP address
- MAC address
- DNS server
- DHCP enabled/disabled
- lease information

**Purpose:**

Used when more detailed network configuration is needed.

---

## 3. ping

This command checks whether another device or website is reachable.

**Command:**

ping google.com

or

ping 192.168.1.1

**Purpose:**

- test connectivity
- check whether network is working
- measure response time

**Example:**

If reply comes, the destination is reachable.

---

## 4. tracert

This command shows the path taken by packets from source to destination.

**Command:**

tracert google.com

**Purpose:**

- find route path
- locate delay or failure point in network

---

## 5. nslookup

This command finds the IP address of a domain name.

**Command:**

nslookup google.com

**Purpose:**

- test DNS working
- convert domain name to IP address

---

## 6. arp -a

This command shows the ARP table.

**Command:**

arp -a

**Purpose:**

- see IP address and MAC address mapping

- troubleshoot LAN communication

---

### 7. netstat

This command shows active network connections and listening ports.

**Command:**

netstat

**Common form:**

netstat -an

**Purpose:**

- see open ports
- see active connections
- troubleshoot services and connections

---

### 8. hostname

This command shows the computer name.

**Command:**

hostname

**Purpose:**

Used to know the host/computer name in the network.

---

### 9. getmac

This command shows the MAC address of the system.

**Command:**

getmac

**Purpose:**

Used to identify physical network address.

---

### 10. pathping

This command combines features of ping and tracert.

**Command:**

pathping google.com

**Purpose:**

- check packet loss
- analyze route performance

---

## B. Basic Networking Commands in Linux

.........................................killerkoda.com

### 1. ifconfig

This command shows or configures network interface information.
In some modern Linux systems, ifconfig may not be installed by default.

**Command:**

ifconfig

**Purpose:**

- see IP address
- check network interface details

---

### 2. ip a

This is the modern Linux command for viewing IP address information.

**Command:**

ip a

or

ip addr

**Purpose:**

- show IP address
- show interface status
- display network interface details

---

**3. ping**

Same as Windows, used to test connectivity.

**Command:**

ping google.com

**Note:**

In Linux, ping runs continuously until stopped with Ctrl + C.

**Purpose:**

- test reachability
- test network response

---

**4. traceroute**

This command shows the route packets take to reach destination.

**Command:**

traceroute google.com

**Purpose:**

- view path
- detect routing problem

---

**5. nslookup**

Used to find DNS information.

**Command:**

nslookup google.com

**Purpose:**

- test DNS resolution
- find IP from domain name

---

**6. dig**

This is another DNS lookup command in Linux.

**Command:**

dig google.com

**Purpose:**

- get detailed DNS information
- query DNS records

---

**7. arp**

Shows ARP table.

**Command:**

arp -a

**Purpose:**

- display IP to MAC mapping

---

**8. netstat**

Shows connections, routes, interface statistics.

**Command:**

netstat -an

**Purpose:**

- display active connections

- show listing ports

---

**9. ss**

This is a modern replacement for netstat in Linux.

**Command:**

ss -tuln

**Purpose:**

- show open ports
- show socket information

---

**10. hostname**

Shows the system hostname.

**Command:**

hostname

**Purpose:**

Used to identify system name.

---

**11. route**

Displays routing table.

**Command:**

route -n

**Purpose:**

- view routing path
- troubleshoot routing issue

---

**12. ip route**

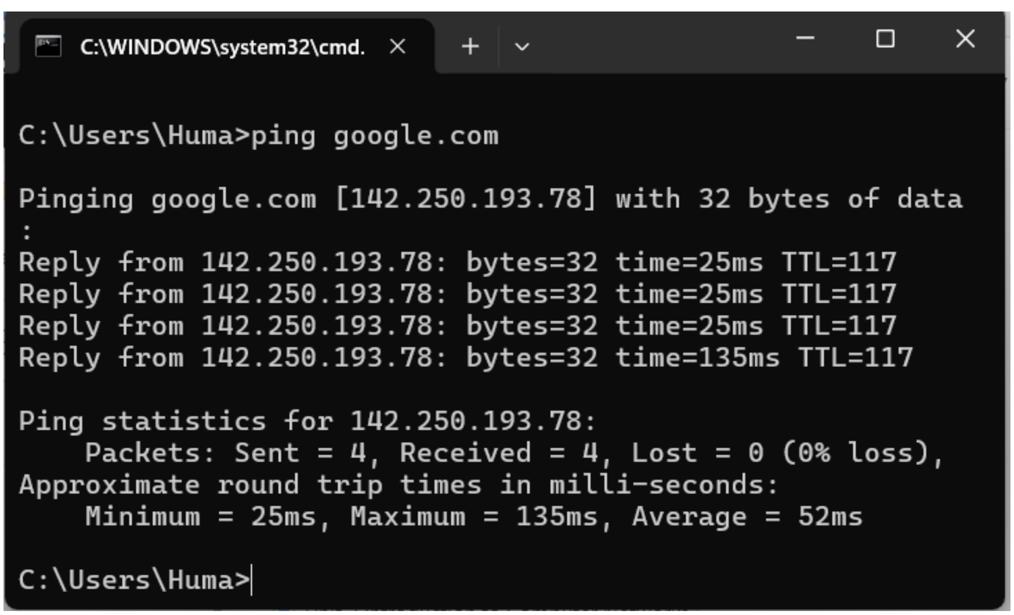
Modern command to show routing table.

**Command:**

ip route

**Purpose:**

- display route entries
- check gateway and route path




**Sanjeev Thapa, Er. DevOps, SRE, CKA, RHCSA, RHCE, RHCSA-Openstack, MTCNA, MTCTCE, UBSRS, HEv6, Research Evangelist**